QTM 220: Regression Analysis

Emory University

	Lecture room:	Math and Science Center E208
	Lecture time:	Tuesdays and Thursdays, 2:30-3:45pm
	Lab room:	PAIS 230
	Lab time:	Fridays, 2:30-3:20pm
Instructors:	Adam Glynn	Allison Cuttner
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Office:	PAIS	PAIS
Office hours:		
	TA:	
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Course Description

(from previous syllabus) This course introduces students to widely-used procedures for regression analysis and provides intuitive, applied, and formal foundations for more advanced methods treated in later courses. The course covers descriptive, population and causal inference with regression from a modern perspective. While the course will emphasize the mathematical foundations of these concepts, each topic will also cover the implementation of the relevant methods in an application through the use of the statistical program R.

This course represents the culmination of the required courses for QTM majors. It uses the coding skills learned in QTM 150 and the math skill learned in Math 210 and QTM 220 to conduct much of the analysis discussed in QTM 110. QTM 220 also provides an introduction and a foundation for future study of topics such as machine learning and causal inference.

new things to thing about: How does this course fit in to the curriculum/sequence for the rest of the major? Why is it important for students to devote a lot of time/effort to? Why are description, population inference, and causal inference separated? Why are different types of covariates separated? How will this course prepare students to do 'cool' stuff like causal inference and machine learning?

Course Goals By the end of this course, students will be able to

These are copied from old syllabus and need to be updated

- Be able to conduct a number of regression analyses and correctly interpret the results based on research design concepts learned in QTM 110.
- Understand the properties of regression estimators so as to understand when these estimators are reliable.

- Distinguish between descriptive, population, and causal interpretations of regression findings, and understand the assumptions that are necessary for each.

new ideas:
student will be able to...

- formalize questions about the world (descriptive, population, and causal)
- use regression to answer those questions
- quantify uncertainty and error
- understand when certain techniques are appropriate to use (or not)

What other tools/skills (including specific R skills) will we be confident that students will have mastered by the end of the course?

What tools/skills/concepts will students be exposed to (but not master) by the end of the course?

Prerequisites QTM 210 or equivalent is prerequisite.

Course Requirements

Attendance Lecture attendance is expected. If a student cannot attend lecture, they must email the instructor to explain their absence. Please do not come to class (lecture or lab) if sick.

Problem Sets Problem sets will be assigned (approximately) biweekly throughout the semester. The homework assignments will consist of analytical problems, computer work, and/or data analysis. The homework write-up must be word processed in some manner (exact manner to be determined by the TAs). The due date and time for all homework will be indicated on each assignment and should be submitted through Canvas. No late work will be accepted without an approved excuse.

We encourage students to confer with each other on the homework assignments, but you must write your own solutions (this includes computer work). Copying and pasting code from other students is not allowed, and you must be able to explain anything you turn in.

Midterm Exams The midterms will be in-class exams *There is no collaboration allowed on the midterms.*

Final Exam The final will be held during the assigned exam time and location. *There is no collaboration allowed on the final.*

Lab Session Lab sessions will be run by the TAs. The purpose of the lab sessions is to learn the R programming and data analysis skills useful for regression and other empirical analysis. Students must bring a laptop, with R installed and updated on it, to these sessions. Lab sessions will prepare students to complete the problem sets and review the solutions of previous homeworks. Lab participation and course performance are highly correlated.

Grading

- Problem Sets (35%)
- Midterm Exams (40%)
- Final Exam (25%)

Incomplete Grades

No incomplete grades will be given unless there is an agreement between the instructor and the student **prior** to the end of the course. The instructor retains the right to determine legitimate reasons for an incomplete grade.

Honor Code

All students enrolled at Emory are expected to abide by the Emory College Honor Code. Any type of academic misconduct is not allowed which includes 1) receiving or giving information about the content or conduct of an examination knowing that the release of such information is not allowed and 2) plagiarizing, whether intentionally or unintentionally, in any assignment. For the activities that are considered to be academically dishonest, refer to the Honor Code: http://catalog.college.emory.edu/academic/policies-regulations/honor-code.html.

Accessibility and Accommodations

If you are seeking classroom accommodations or academic adjustments under the Americans with Disabilities Act, you are required to register with Office of Accessibility Services (http://accessibility.emory.edu). To receive academic accommodations for this class, please obtain the relevant letter and meet with the instructor at the beginning of the semester. Students are expected to give two weeks notice of the need for accommodations.

Schedule

Part 1

Lecture 1 (Aug 23 — Aug 24)

Descriptive statistic and binary covariates
Syllabus

Lecture 2 (Aug 28 — Aug 29)

Random sampling and sampling distributions

Lecture 3 (Aug 30 — Aug 31)

Sampling distributions, normal approximations, and confidence intervals

Lecture 4 (Sept 6 — Sept 5)

Causality and Prediction

Lecture 5 (Sept 11 — Sept 7)

Descriptions with Discrete Covariates (beyond binary)

Lecture 6 (Sept 13 — Sept 12)

Misspecification and inference

Lecture 7 (Sept 18 — Sept 14)

Causality

Review (Sept 20 — Sept 19)

Exam (Sept 25 — Sept 21)

Part 2

Lecture 8 (Sept 27 — Sept 26)
Beyond Bins 1: single ordered covariate

Lecture 9 (Oct 2 — Sept 28)

Beyond Bins 2: single ordered covariate

Lecture 10 (Oct 4 — Oct 3)

single ordered covariate, population inference 1

Fall Break (Oct 9 — Oct 10)

Lecture 11 (Oct 11 — Oct 5)

single ordered covariate, population inference 2

Lecture 12 (Oct 16 — Oct 12)

Calculations in continuum

Review (Oct 18 — Oct 17)

Exam (Oct 23 — Oct 19)

Part 3

Lecture 13 (Oct 25 — Oct 24)

multivariate discrete interactions

Lecture 14 (Oct 30 — Oct 26)

Multivariate (X 1D continuous + D discrete)

Lecture 15 (Nov 1 — Oct 31)

Multivariate (X 1D continuous + D continuous)

Lecture 16 (Nov 6 — Nov 2)

Multivariate (X arbitrary + D binary/continous)

Election Day (Nov 8 — Nov 7)

Lecture 17 (Nov 13 — Nov 9)

Population Inference for Interactions 1

Lecture 18 (Nov 15 — Nov 14)

Population Inference for Interactions 2

Lecture 19. (Nov 20 — Nov 16)

Population Inference for Interactions 3

Thanksgiving Break (Nov 22 — Nov 23)

Lecture 20 (Nov 27 — Nov 21)

Population Inference for Interactions 4

Lecture 21 (Nov 29 — Nov 28)

Multivariate causal inference

Lecture 22 (Dec 4 — Nov 30)

Catch up

Review (Off-Schedule — Dec 5)

Exam Period Dec 7 3-5:30pm

Final Exam